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Lecture 7 Scribe

Expectation, CDFs, PDFs and Problem Solving

Course: CSE400 – Fundamentals of Probability in Computing

Instructor: Dhaval Patel, PhD

Ahmedabad University

1. Introduction

This lecture introduces three important concepts in probability theory:

1. Cumulative Distribution Function (CDF)
2. Probability Density Function (PDF)
3. Expectation of Random Variables

These concepts are fundamental for analyzing **random variables and their distributions**, and they are widely used in **computer science, machine learning, statistics, and probabilistic modeling**.

2. Intuition of CDF and PDF (Water Tank Analogy)

To understand the relationship between CDF and PDF, a **water tank analogy** can be used.

Idea

Imagine a **vertical water tank**:

- The **height of the water (h)** represents the **value of a random variable**.
- The **volume of water up to height h**, denoted by $V(h)$, represents the **cumulative probability up to that value**.

Thus,

- **Height $\rightarrow$ Value of random variable**
- **Volume up to that height $\rightarrow$ Cumulative probability**

This idea leads to the **Cumulative Distribution Function (CDF)**.

The **rate at which the volume increases with height** corresponds to the **Probability Density Function (PDF)**.

Mathematical Representation

For a cylindrical tank with radius R :

$$V(h) = \int_0^h \pi R^2 dh$$

which gives

$$V(h) = (\pi R^2)h$$

Here:

- πR^2 behaves like a **constant density**
- This is analogous to the **PDF of a uniform distribution**

Thus:

- PDF → rate of change of probability
 - CDF → accumulated probability
-

3. Cumulative Distribution Function (CDF)

Definition

The **Cumulative Distribution Function (CDF)** of a random variable X is defined as

$$F_X(x) = P(X \leq x)$$

where

$$-\infty < x < \infty$$

It represents the **probability that the random variable takes a value less than or equal to x** .

Interpretation

The CDF gives the **total probability accumulated up to value x** .

If we know the CDF, we know **almost everything about the random experiment described by the random variable X** .

Properties of CDF

A valid CDF must satisfy the following properties:

1. Non-decreasing

$$F_X(x_1) \leq F_X(x_2) \quad \text{if } x_1 < x_2$$

As x increases, probability cannot decrease.

2. Limits at infinity

$$\lim_{x \rightarrow -\infty} F_X(x) = 0$$

$$\lim_{x \rightarrow \infty} F_X(x) = 1$$

This means:

- Probability starts at 0
 - Eventually reaches 1
-

3. Right continuous

The CDF is **continuous from the right**.

4. Probability from CDF

Probability between two values can be obtained using:

$$P(a < X \leq b) = F_X(b) - F_X(a)$$

4. Probability Density Function (PDF)

Definition

For a **continuous random variable**, the **Probability Density Function (PDF)** describes the **density of probability around a value**.

It is defined as the derivative of the CDF.

$$f_X(x) = \frac{d}{dx} F_X(x)$$

Relationship between CDF and PDF

$$F_X(x) = \int_{-\infty}^x f_X(t) dt$$

Thus:

- PDF $\rightarrow$ derivative of CDF
 - CDF $\rightarrow$ integral of PDF
-

Properties of PDF

A function $f_X(x)$ is a valid PDF if it satisfies:

1. Non-negativity

$$f_X(x) \geq 0$$

for all x .

2. Total probability equals 1

$$\int_{-\infty}^{\infty} f_X(x) dx = 1$$

3. Probability between two values

$$P(a < X < b) = \int_a^b f_X(x) dx$$

5. Expectation of Random Variables

Definition

The **expectation (or expected value)** of a random variable is the **average value** the variable takes over many repetitions of the experiment.

It represents the **mean value of the probability distribution**.

Expectation of a Discrete Random Variable

If X takes values x_i with probability p_i ,

$$E[X] = \sum x_i p_i$$

Expectation of a Continuous Random Variable

If X has PDF $f_X(x)$,

...

$$E[X] = \int_{-\infty}^{\infty} xf_X(x)dx$$

Example of Expectation

Suppose a random variable takes values with different probabilities.

To compute expectation:

1. Multiply each value with its probability
2. Add all the results

$$E[X] = \sum x_i P(X = x_i)$$

This gives the **long-run average value**.

6. Expectation of a Function of a Random Variable

Sometimes we want the expectation of a **function of the random variable**, such as:

$$E[g(X)]$$

Discrete case

$$E[g(X)] = \sum g(x_i)P(X = x_i)$$

Continuous case

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f_X(x)dx$$

7. Linear Operations with Expectation

Expectation follows **linearity property**.

For constants a and b :

$$E[aX + b] = aE[X] + b$$

More generally:

$$E[aX + bY] = aE[X] + bE[Y]$$

Important points:

- Linearity holds **even if variables are not independent**
 - Makes expectation calculations much easier
-

8. Moments of Random Variables

Moments describe **characteristics of a distribution**.

nth Moment

The **nth moment about the origin** is

$$E[X^n]$$

Moments provide information about:

- mean
 - spread
 - shape
-

9. Central Moments

Central moments are moments **around the mean**.

$$E[(X - \mu)^n]$$

where

$$\mu = E[X]$$

10. Variance

Variance measures the **spread of the distribution**.

$$\text{Var}(X) = E[(X - \mu)^2]$$

Alternative formula:

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

Interpretation

- Large variance → values spread widely
 - Small variance → values close to the mean
-

11. Skewness

Skewness measures **asymmetry of the distribution**.

- **Positive skew** → long right tail
 - **Negative skew** → long left tail
-

12. Kurtosis

Kurtosis measures **how heavy the tails of a distribution are**.

- High kurtosis → heavy tails
- Low kurtosis → lighter tails

It describes **peakedness of the distribution**.

13. Summary of Lecture 7

This lecture covered key probability concepts:

CDF

- $F_X(x) = P(X \leq x)$
- Non-decreasing
- Limits from 0 to 1

PDF

- $f_X(x) = \frac{d}{dx}F_X(x)$
- Integral equals 1

Expectation

- Average value of random variable
- Discrete: $E[X] = \sum x_i p_i$
- Continuous: $E[X] = \int x f(x) dx$

Expectation Properties

- Linearity
- Expectation of functions

Moments

- nth moment
- Variance
- Skewness
- Kurtosis

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- Convert this **lecture scribe** into a clean PDF or LaTeX notes for printing
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CSE400 – Fundamentals of Probability in Computing

Lecture 7: Expectation, CDFs, PDFs and Problem Solving

Instructor: Dhaval Patel, PhD

SEAS – Ahmedabad University

Date: 27 January 2025

1. Overview of the Lecture

This lecture discusses important concepts used in probability theory and statistical modeling:

1. Cumulative Distribution Function (CDF)
2. Probability Density Function (PDF)
3. Expectation of Random Variables
4. Expectation of a Function of a Random Variable
5. Linear Operations with Expectation
6. Moments and Central Moments
7. Variance, Skewness and Kurtosis

These concepts help describe **the behavior, average value, and distribution of random variables**.

2. Cumulative Distribution Function (CDF)

Definition

The **Cumulative Distribution Function (CDF)** of a random variable X is defined as:

$$F_X(x) = P(X \leq x)$$

It represents the **probability** that the random variable takes a value less than or equal to X .

Interpretation

The CDF measures **accumulated probability** up to a certain value.

If we know the CDF of a random variable, we can determine the **probability for any range of values**.

Properties of the CDF

The CDF satisfies the following properties:

1. Non-Decreasing Property

The CDF is a **monotonically increasing function**.

$$F_X(x_1) \leq F_X(x_2) \quad \text{if } x_1 < x_2$$

This means probability cannot decrease as X increases.

2. Limits at Infinity

$$\lim_{x \rightarrow -\infty} F_X(x) = 0$$

$$\lim_{x \rightarrow \infty} F_X(x) = 1$$

Thus the cumulative probability starts at **0** and approaches **1**.

3. Right Continuity

The CDF is **right-continuous**, meaning:

$$\lim_{h \rightarrow 0^+} F_X(x + h) = F_X(x)$$

4. Probability Between Two Values

Using the CDF we can compute probability in an interval:

$$P(a < X \leq b) = F_X(b) - F_X(a)$$

3. Example Concept for CDF

The CDF describes how probability **accumulates across possible values of the random variable**.

For example:

If we know $F_X(x)$, then the probability that the variable lies between two points is simply the **difference between the CDF values**.

This property makes the CDF **very useful in probability analysis and modeling**.

4. Probability Density Function (PDF)

Definition

For a **continuous random variable**, the **Probability Density Function (PDF)** describes the **relative likelihood of the variable taking a particular value**.

The PDF is defined as the derivative of the CDF:

$$f_X(x) = \frac{d}{dx} F_X(x)$$

Relationship Between PDF and CDF

The CDF can be obtained from the PDF using integration:

$$F_X(x) = \int_{-\infty}^x f_X(t) dt$$

Thus:

- PDF $\rightarrow$ derivative of CDF
 - CDF $\rightarrow$ integral of PDF
-

Properties of the PDF

A function $f_X(x)$ is a valid PDF if it satisfies the following properties:

1. Non-negativity

$$f_X(x) \geq 0$$

for all x .

2. Total Probability Equals 1

$$\int_{-\infty}^{\infty} f_X(x) dx = 1$$

3. Probability Between Two Points

The probability that the random variable lies between a and b is

$$P(a < X < b) = \int_a^b f_X(x) dx$$

5. Expectation of Random Variables

Definition

The **Expectation (Expected Value)** of a random variable represents its **average value over many repeated experiments**.

It is the **probability-weighted mean** of the possible values of the random variable.

Expectation of a Discrete Random Variable

If a random variable X takes values x_i with probabilities p_i , then

$$E[X] = \sum x_i p_i$$

Expectation of a Continuous Random Variable

If X has PDF $f_X(x)$, then

$$E[X] = \int_{-\infty}^{\infty} xf_X(x) dx$$

Example Concept

The expectation gives the **long-run average value** of the random variable.

If an experiment is repeated many times, the **average outcome approaches the expected value**.

6. Expectation of a Function of a Random Variable

Sometimes we need the expectation of a function of a random variable, denoted as:

$$E[g(X)]$$

Discrete Case

$$E[g(X)] = \sum g(x_i)P(X = x_i)$$

Continuous Case

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f_X(x) dx$$

7. Linear Operations with Expectation

Expectation satisfies the **linearity property**.

For constants a and b :

$$E[aX + b] = aE[X] + b$$

For two random variables X and Y :

$$E[aX + bY] = aE[X] + bE[Y]$$

Important point:

Linearity holds **even when the random variables are not independent**.

8. Moments of Random Variables

Moments describe important **characteristics of the probability distribution**.

nth Moment

The **nth moment about the origin** is defined as:

$$E[X^n]$$

Moments provide information about:

- mean
 - spread
 - shape of the distribution
-

9. Central Moments

Central moments are calculated **around the mean** of the random variable.

$$E[(X - \mu)^n]$$

where

$$\mu = E[X]$$

These moments describe the **structure of the distribution relative to its mean**.

10. Variance

Variance measures the **dispersion or spread of the distribution**.

$$\text{Var}(X) = E[(X - \mu)^2]$$

An equivalent formula is:

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

Interpretation

- **Large variance** → data widely spread
 - **Small variance** → data clustered near the mean
-

11. Skewness

Skewness measures the **asymmetry of a probability distribution**.

Types of skewness:

1. **Positive skewness**
 - Distribution tail extends toward the right.
2. **Negative skewness**
 - Distribution tail extends toward the left.

Skewness helps identify **bias in the distribution shape**.

12. Kurtosis

Kurtosis measures the **peakedness and tail heaviness** of a distribution.

- **High kurtosis** → heavy tails and sharp peak
- **Low kurtosis** → flatter distribution

Kurtosis provides insight into **the probability of extreme values**.

13. Summary of Lecture 7

The lecture introduced key concepts used to describe random variables.

Key Points

CDF

$$F_X(x) = P(X \leq x)$$

- cumulative probability

- non-decreasing
 - limits between 0 and 1
-

PDF

$$f_X(x) = \frac{d}{dx}F_X(x)$$

- describes probability density
 - integrates to 1
-

Expectation

$$E[X] = \sum x_i p_i$$

or

$$E[X] = \int x f_X(x) dx$$

- represents long-run average value
-

Additional Concepts

- Expectation of a function $E[g(X)]$
- Linearity of expectation
- Moments and central moments
- Variance
- Skewness
- Kurtosis

These tools are fundamental for **analyzing probability distributions and solving statistical problems in computing.**

✓ If you want, I can also:

- Convert this into **clean LaTeX notes (perfect for exam revision)**
- Make a **3–4 page condensed exam cheat sheet from Lecture 7**

- Or extract **all solved problems** from the slides with **step-by-step solutions**.



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CSE400 – Fundamentals of Probability in Computing

Lecture 7: Expectation, CDFs, PDFs and Problem Solving

Instructor: Dhaval Patel, PhD

SEAS – Ahmedabad University

Date: 27 January 2025

1. Lecture Outline

The lecture covers the following major topics:

1. Cumulative Distribution Function (CDF)

- Definition
- Properties
- Example

2. Probability Density Function (PDF)

- Definition
- Properties
- Example

3. Expectation of Random Variables (RVs)

- Definition
- Example

4. Expectation of a Function of Random Variables

5. Linear Operation with Expectation

6. Moments and Central Moments of Random Variables

- Variance
 - Skewness
 - Kurtosis
-

2. Random Variables (Brief Review)

A **Random Variable (RV)** is a function that assigns a numerical value to each outcome of a random experiment.

Random variables are of two types:

- 1. Discrete Random Variable**
- 2. Continuous Random Variable**

Probability distributions describe how probabilities are assigned to these values.

3. Cumulative Distribution Function (CDF)

Definition

The **Cumulative Distribution Function (CDF)** of a random variable X is defined as

$$F_X(x) = P(X \leq x)$$

It represents the **probability** that the random variable takes a value less than or equal to X .

Interpretation

The CDF gives the **total accumulated probability** up to a particular value of the random variable.

If the CDF of a random variable is known, we can determine **probabilities for any interval of values**.

4. Properties of CDF

The CDF must satisfy the following properties.

1. Non-Decreasing Property

The CDF is a **monotonically increasing function**.

$$F_X(x_1) \leq F_X(x_2) \quad \text{if } x_1 < x_2$$

This means probability never decreases as X increases.

2. Boundary Conditions

$$\lim_{x \rightarrow -\infty} F_X(x) = 0$$

$$\lim_{x \rightarrow \infty} F_X(x) = 1$$

Thus:

- At very small values, cumulative probability approaches **0**
 - At very large values, cumulative probability approaches **1**
-

3. Right Continuity

The CDF is **right-continuous**, meaning

$$F_X(x) = \lim_{h \rightarrow 0^+} F_X(x + h)$$

4. Probability Using CDF

The probability that the random variable lies between two values is

$$P(a < X \leq b) = F_X(b) - F_X(a)$$

5. Example of CDF

If the CDF of a random variable is known, then the probability of any interval can be determined by subtracting CDF values.

For example,

$$P(a < X \leq b) = F_X(b) - F_X(a)$$

Thus, the CDF completely describes the probability distribution.

6. Probability Density Function (PDF)

Definition

For a **continuous random variable**, the **Probability Density Function (PDF)** represents the **density of probability at each value**.

The PDF is defined as the derivative of the CDF:

$$f_X(x) = \frac{d}{dx} F_X(x)$$

Relationship Between PDF and CDF

$$F_X(x) = \int_{-\infty}^x f_X(t) dt$$

Thus:

- CDF is the integral of PDF
 - PDF is the derivative of CDF
-

7. Properties of PDF

A function $f_X(x)$ is a valid probability density function if it satisfies the following properties.

1. Non-Negativity

$$f_X(x) \geq 0$$

for all x .

2. Total Probability Equals 1

$$\int_{-\infty}^{\infty} f_X(x) dx = 1$$

3. Probability Between Two Values

$$P(a < X < b) = \int_a^b f_X(x) dx$$

8. Expectation of Random Variables

Definition

The **Expectation (Expected Value)** of a random variable represents the **average value obtained** if the experiment is repeated many times.

It is the **probability-weighted mean** of the values taken by the random variable.

Expectation for Discrete Random Variables

If a random variable X takes values $x_1, x_2, \dots, x_n$ with probabilities $p_1, p_2, \dots, p_n$,

$$E[X] = \sum_{i=1}^n x_i p_i$$

Expectation for Continuous Random Variables

If X has PDF $f_X(x)$,

$$E[X] = \int_{-\infty}^{\infty} xf_X(x)dx$$

9. Example of Expectation

To compute expectation:

1. Multiply each value of the random variable by its probability.
2. Sum all these values.

This gives the **long-run average value of the random variable**.

10. Expectation of a Function of a Random Variable

Sometimes we want the expectation of a function of the random variable.

Let $g(X)$ be a function of X .

Discrete Case

$$E[g(X)] = \sum g(x_i)P(X = x_i)$$

Continuous Case

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f_X(x)dx$$

11. Linear Operations with Expectation

Expectation follows the **linearity property**.

For constants a and b :

$$E[aX + b] = aE[X] + b$$

For two random variables X and Y :

$$E[aX + bY] = aE[X] + bE[Y]$$

Important fact:

Linearity holds **even if X and Y are not independent.**

12. Moments of Random Variables

Moments describe **important characteristics of a distribution.**

nth Moment

The **nth moment about the origin** is

$$E[X^n]$$

Moments help describe:

- mean
 - shape
 - spread of the distribution.
-

13. Central Moments

Central moments measure moments **around the mean.**

$$E[(X - \mu)^n]$$

where

$$\mu = E[X]$$

14. Variance

Variance measures **how far values of the random variable spread from the mean.**

$$\text{Var}(X) = E[(X - \mu)^2]$$

Alternative formula:

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

15. Skewness

Skewness measures the **asymmetry of a probability distribution**.

Types:

- **Positive Skew** → distribution tail extends to the right
 - **Negative Skew** → distribution tail extends to the left
-

16. Kurtosis

Kurtosis measures the **peakedness and heaviness of the tails of a distribution**.

- High kurtosis → sharper peak and heavier tails
 - Low kurtosis → flatter distribution
-

17. Final Summary of Lecture 7

Main concepts discussed in this lecture include:

CDF

$$F_X(x) = P(X \leq x)$$

- cumulative probability function
-

PDF

$$f_X(x) = \frac{d}{dx} F_X(x)$$

- describes probability density
-

Expectation

$$E[X] = \sum x_i p_i$$

or

$$E[X] = \int x f_X(x) dx$$

- long-run average value
-

Additional Statistical Measures

- Expectation of function $E[g(X)]$
- Linearity of expectation
- Moments and central moments
- Variance
- Skewness
- Kurtosis

These tools are fundamental for **probability modeling, statistical analysis, and computing applications.**

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- Converting this into **clean LaTeX notes exactly formatted like a lecture scribe** (very useful for submission).
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